

Block completion at $L = 0.3$ fails:
the $20\times$ drop is an $n = 48$ artefact,
jets $\geq \psi_{10}$ are garbage, QR-IBP kills S
PRIME.RDAGGER.BLOCK_COMPLETION.01 — round 484

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Abstract

Round 483 sealed Filon on $[0, 150]$ and a four-term IBP tail, and Higham-certified the finite even Schur ($\mu \geq 1.20 \cdot 10^{-3}$). The SATZ $\lambda_*(0.3) \geq c$ was blocked by leftover $\|P_{\text{block}}AP_{\geq 12}\|_F = 2.76 \cdot 10^{-2}$ against budget $2.4 \cdot 10^{-4}$. The hoped close was a $\sim 20\times$ coupling drop per even double-step, so that a block through ψ_{18} would push leftover below budget, together with a Nyström- $n = 128$ (or QR-polynomial) certificate of the high-mode jets. Both routes fail. A mode is usable iff $|u(L)|$ at n and $2n$ agree to 10^{-3} : $\psi_0, \psi_2, \psi_4, \psi_6$ pass; ψ_8 is marginal (10^{-4}); ψ_{10} and above are $O(1)$ -unstable at every $n \leq 128$ (λ_{10} sits at the binary64 floor). On the resolved $n = 96$ surface the NO-TAIL leftover after ψ_{10} is $1.38 \cdot 10^{-1}$ (the $n = 48$ figure $2.76 \cdot 10^{-2}$ was an undercount); after ψ_{18} it is $9.60 \cdot 10^{-2}$; after ψ_{20} , $4.40 \cdot 10^{-2}$ (factor ~ 180 versus budget). The drop $10 \rightarrow 18$ is $1.43\times$, not $20\times$. Even Legendres QR-orthogonalised against certified $\psi_0.. \psi_8$ leave leftover $O(0.08)$ at 12 extra columns; four-term sinc-IBP on those vectors makes $\lambda_{\min}(S) \approx -0.065$. Enlarging B to $\{\psi_0.. \psi_8\}$ recovers $\|C_{\text{rest}}\| \approx 0.29$, the r480 κ_{high} , still alive in the discrete nullspace. Verdict **STUCK(20x-drop-falsified@n96; jets>=10-garbage@n128; QR-IBP-kills-S)**. Fourier-HS is not reused. Yoshida-Bombieri zone. No RH claim.

Status. **STUCK(20x-drop-falsified@n96; jets>=10-garbage@n128; QR-IBP-kills-S)**. No promotion of an RH claim.

Claim boundary. Nothing here is $\lambda_*(0.3) \geq c$. Block completion does not close the r483 leftover. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $Q_W = A - P + \Pi$, prime nodes $n \leq e^{2L}$. At $L = 0.3$ one has $P \equiv 0$ (scramble-sensitive, vacuum). The r469/r479–r483 contract is in force: $t_* = 8$, $\varepsilon = \lambda_6(K_8) \leq 1.46 \cdot 10^{-8}$, $c_{\text{tail}} \geq 0.2408$, Filon $[0, 150]$ and four-term IBP as in r483, even block $\{\psi_0, \psi_2, \psi_4\}$.

2 Garbage criterion

A Nyström interpolant is used in IBP only if the endpoint jet $|u(L)|$ agrees at n and $2n$ to 10^{-3} (the IBP entry budget).

Mode	$\lambda(K_8)$	$ u(L) _{48}$ vs 64	96 vs 128	Status
ψ_0, ψ_2, ψ_4	0.935, 0.082, $9.7 \cdot 10^{-5}$	10^{-13}	10^{-13}	certified
ψ_6	$1.46 \cdot 10^{-8}$	$1.6 \cdot 10^{-9}$	10^{-9}	certified
ψ_8	$6.16 \cdot 10^{-13}$	$1.4 \cdot 10^{-4}$	$1.6 \cdot 10^{-4}$	marginal
ψ_{10+}	$\leq 10^{-16}$ (floor)	$O(1)$	$O(1)$	garbage

True $\lambda_{10}(K_8)$ is theoretically far below 10^{-16} . Discrete “modes” 10, 12, ... are an ONB of the Nyström nullspace, not Slepians. They are not used as a pad.

3 The 20× drop is false

NO-TAIL leftover $\|P_{\{0,2,4\}}AP_{\text{rest}}\|_F$ on $n = 96$, $T = 150$:

C through	leftover F	vs. budget $2.4 \cdot 10^{-4}$
ψ_8	$1.83 \cdot 10^{-1}$	760×
ψ_{10}	$1.38 \cdot 10^{-1}$	575×
ψ_{18}	$9.60 \cdot 10^{-2}$	400×
ψ_{20}	$4.40 \cdot 10^{-2}$	183×

Drop $10 \rightarrow 18$ is 1.43×. The r483 number $2.76 \cdot 10^{-2}$ after ψ_{10} was an $n = 48$ undercount (only columns 12, 14 exist before junk). It is not a SATZ pad.

4 QR escape

Any orthonormal extension of certified $\{\psi_0, \psi_2, \psi_4, \psi_6, \psi_8\}$ is legal for the Schur; $\varepsilon = \|(1-P)K(1-P)\|$ need only be controlled at the end. Even Legendres $P_0, P_2, \dots, P_{28}$, QR-orthogonalised on the $n = 96$ nodes, give a 20-vector ONB. NO-TAIL leftover after twelve extra columns is $8.0 \cdot 10^{-2}$ (completeness leak, same species as r481 even-Legendre). Four-term IBP via the sinc extension of those QR samples (the kernel interpolant is not the polynomial) yields $\lambda_{\min}(S) \approx -0.065$. Not used as a pad.

Enlarging B to $\{\psi_0.. \psi_8\}$ leaves operator leftover 0.294, i.e. r480 κ_{high} , still present in the discrete complement.

5 Error budget

Source	Size	vs. r483 Higham μ
Filon / IBP on certified $\psi_0.. \psi_8$	as r483, $\mu \geq 1.20 \cdot 10^{-3}$	—
Would-be c_* (leftover = 0)	$1.68 \cdot 10^{-4}$	—
$n = 48$ leftover after ψ_{10} (r483)	$2.76 \cdot 10^{-2}$	undercount
$n = 96$ leftover after ψ_{20}	$4.40 \cdot 10^{-2}$	183× budget
QR leftover after 12 columns	$8.0 \cdot 10^{-2}$	330× budget
Enlarge- B leftover ($= \kappa_{\text{high}}$)	0.294	lives
QR sinc-IBP $\lambda_{\min}(S)$	-0.065	kills S

Exact deficit: leftover after ψ_{20} on $n = 96$ is $4.40 \cdot 10^{-2}$ against budget $2.4 \cdot 10^{-4}$ (factor 183). The honest L^2 leftover is the discrete-complement coupling ≈ 0.29 .

6 Kills

Object	Round	Status
Fourier-HS majorant	r478	not reused
Crude κ_{high}	r480	killed; lives in complement
Integer Fourier ONB	r481	$Q_{0n} \sim 1/n$
σ^2 as $\ Au\ _{[-L,L]}$	r481	killed
IBP envelope as pad	r481	killed
Exp-sum Ci to ∞	r483	killed
$n = 48$ leftover $2.76 \cdot 10^{-2}$ as SATZ pad	r484	undercount
$20\times$ drop / block through ψ_{18}	r484	falsified
High-mode jets $n = 128$	r484	garbage $\geq \psi_{10}$
QR sinc-IBP tail	r484	$\lambda_{\min}(S) < 0$

7 Scaling at $L = 0.8$

The exponential-sum / Filon / IBP machine survives (P is a cosine sum; scramble becomes live). Block completion does not become easier: the leftover is a completeness statement about C , not about T or about L . A new handle would have to bound $\|P_{\text{block}}A(1 - P_V)\|$ without representing the complement by garbage eigenmodes.

8 What is not claimed

No $\lambda_*(0.3) \geq c$. The finite r483 Schur remains interval-PD; the leftover C is not smaller than r483 claimed, it is larger once the Nyström surface is resolved. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.